

Tree Health Analyzer

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Abstract—The Tree Health Analyzer is a software application developed in Java aimed at assessing and monitoring the health of trees in various environments. As trees are vital components of our ecosystem, their health directly impacts biodiversity, air quality, and urban aesthetics. This application provides a user-friendly interface for arborists, environmentalists, and urban planners to input and analyze key health indicators such as leaf density, soil moisture, and species type. Utilizing a simple yet effective algorithm, the Tree Health Analyzer evaluates the collected data to classify tree health into categories such as "Healthy," "Stressed," or "Diseased." The application leverages Java's robust capabilities to ensure efficient data processing and real-time feedback. Additionally, the graphical user interface (GUI) built with JavaFX enhances user experience, allowing for easy data entry and immediate visualization of health assessments. The Tree Health Analyzer not only facilitates early detection of potential health issues but also supports informed decision-making regarding tree management and conservation strategies. By promoting proactive care, this tool contributes to the sustainability of urban and rural green spaces, ultimately fostering healthier ecosystems. This project exemplifies the integration of technology and environmental stewardship, providing a valuable resource for maintaining the vitality of tree populations for future generations.[1]

Keywords: Tree,Disease,Cotton,Potato,Tomato,Analyse,Corn,
Precautions.Leaves.

I. INTRODUCTION

Trees are essential for ecological balance, offering benefits such as air purification, carbon sequestration, and biodiversity support. However, their health is increasingly threatened by climate change, urbanization, and deforestation. Traditional methods of assessing tree health are labor-intensive and prone to human error. The motivation behind developing a Tree Health Analyzer is to create an efficient and precise monitoring tool using advanced technologies like remote sensing. This will enable timely intervention and management practices to maintain treehealth.[1]

Trees are essential for our ecosystem, providing benefits like carbon sequestration, air purification, and habitat provision. Monitoring tree health is crucial for preserving these benefits. The tree health analyzer aims to detect and diagnose tree health issues early using advanced technologies, ensuring effective treatment and promoting overall tree vitality. The system works by collecting information such as leaf color, bark texture, and growth patterns, which are then processed to determine the tree's overall health. Additionally, it provides guidance on watering, fertilization, and pruning, ensuring optimal care for different tree species. With a user friendly interface, the analyzer simplifies tree maintenance for garden- ers, arborists, and forestry departments.[3]

A Tree Care Analyzer is a software tool designed to help users monitor and maintain the health of trees by providing valuable insights based on environmental factors, tree conditions, and user inputs. Unlike systems that rely on image processing, this analyzer focuses on manual data entry, predefined rules, and sensor-based inputs (if applicable) to assess tree health. It allows users to track tree growth, analyze soil and environmental conditions, diagnose potential diseases, and receive personalized care recommendations.[7]

The system allows users to manually input tree characteristics, including leaf color, bark texture, branch growth, and overall condition. It can also integrate sensor- based data such as soil moisture, temperature, humidity, and pH levels to enhance the analysis. Based on these inputs, Java based algorithms process the information and generate insights, including disease predictions and care suggestions such as watering schedules, fertilization plans, and pruning recommendations.[4]

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users to track tree growth, analyze soil and environmental conditions, diagnose potential diseases, and receive personalized care recommendations.[6]

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II. RELATED WORK

Tree health analysis has gained significant attention in recent years due to its role in environmental conservation and agriculture. Several research studies have explored various techniques for diagnosing tree health, utilizing image processing, machine learning, and IoT-based monitoring systems. Java has been widely used in these implementations due to its platform independence, object-oriented capabilities, and robust ecosystem for integrating AI and data analysis libraries.[4]

One approach in tree health assessment involves the use of Java-based image processing techniques to detect diseases, discoloration, and structural anomalies in tree leaves and bark. Java libraries such as OpenCV and Java Advanced Imaging

(JAI) have been employed to preprocess and analyze images captured by drones or mobile devices. Researchers have also leveraged deep learning models, such as convolutional neural networks (CNNs), which can be integrated into Java applications using frameworks like DeepLearning4J. These models assist in classifying tree diseases based on visual symptoms, enabling early intervention to prevent the spread of infections.

Additionally, IoT-based tree health monitoring systems using Java have been explored in various studies. These systems typically include sensors to measure parameters such as soil moisture, temperature, humidity, and chlorophyll content. Java-based backend applications process the collected data and provide real-time analytics through dashboards. The integration of Java with cloud computing services like AWS and Firebase enables remote monitoring, making tree health analysis more accessible and efficient. Despite the advancements, challenges remain in enhancing the accuracy of tree health analysis models and ensuring scalability for large-scale implementations. Future research could focus on improving Java-based real-time processing capabilities, enhancing dataset quality, and integrating AI-driven automation for better decisionmaking.

Java's powerful computational features, tree health analyzers can continue to evolve into more reliable and efficient tools for environmental sustainability. [8]

TABLE I: Literature Review

Year	Author	Title	Methodology
2022	Rich Hallett,Rachel Holmes,Michelle Johnson, Ben Mertz,	A HEALTHY TREES, HEALTHY CITIES	The "Healthy Trees, Healthy Cities" initiative focuses on urban tree health monitoring and management. Its methodology involves using tools like the HTHC Mobile App to assess tree health through visual signs of stress, pest detection, and health metrics. The process includes conducting health checks to identify non-specific stress indicators, estimating percentages for health metrics, and calculating a stress index. Pest detection checks are also performed to identify potential threats. [1]
2023	Richard Hallett and Tanner Hallett.	Citizen Science and Tree Health Assessment	The methodology described in "Citizen Science and Tree Health Assessment" involves training volunteers to assess tree health using standardized protocols. In the study, the U.S. Forest Service trained a Boy Scout troop in Wisconsin to evaluate ash trees affected by the emerald ash borer. The training included a two-hour session on measuring tree stress symptoms, such as defoliation, leaf discoloration, and fine-twig dieback, using a five-class system. [2]
2022	Lara A. Roman, Bailey C. Smith, Dana Dentice, Mindy Maslin, and Glen Abrams	Monitoring Young Tree Survival with Citizen Scientists	The methodology in "Citizen Science and Tree Health Assessment" involves training volunteers to assess tree health using standardized protocols. In this study, the U.S. Forest Service trained a Boy Scout troop in Wisconsin to evaluate ash trees affected by the emerald ash borer. ... [3]

2023	Hao Youa; Chenxi Zua; Xianfa Mao a; Jiale Suna; Jia Liua; Cheng Luoa; Yi Liua;	Low-Cost Chamber Method for Measuring CO fluxes on the surface of tree trunks Using Recycled Polymer Containers	The methodology described in "Low-Cost Chamber Method for Measuring CO Fluxes on the Surface of Tree Trunks Using Recycled Polymer Containers" involves designing and utilizing chambers made from recycled polymer containers to measure CO fluxes. These chambers are low-cost and environmentally friendly, making them accessible for widespread use. [4]
2024	: Krzyzaniak, M.; Swierk, D.; Antoszewski, P	Tree Health Analyzer	It involves analyzing environmental factors such as air and soil pollution that impact tree health. The study uses dendrometric parameters and health status assessments for selected tree species, including small-leaved limes, maples, oaks, and chestnut trees. Soil samples are tested for nutrient content (P, K, Ca, Mg, Na), while air pollution data is obtained from accessible sources. [5]
2022	Azadeh Abdollahnejad * and Dimitrios Panagiotidis	Tree Species Classification and Health Status Assessment	a methodology for classifying tree species and assessing their health status using unmanned aerial systems (UAS) equipped with multispectral imaging sensors. The research was conducted in a mixed broadleaf-conifer forest near Oplany, Czech Republic, covering a 180 × 160 m area dominated by Norway spruce, Scots pine, Downy oak, Hornbeam, and Silver birch. Field assessments in August 2019 involved visual identification of tree species and health conditions, particularly focusing on bark beetle infestations. Tree positions were precisely recorded using a Trimble M3 total station. [6]
2022	Krisztina Szabo' 1,* , Eszter T' oke 2 and Attila Gergely 3	Evaluating Spherical Trees in the Urban Environment in Budapest	a study to evaluate spherical crown tree species in Budapest's urban environment, focusing on their distribution, growth characteristics, and overall suitability for city landscapes. The research analyzed eight spherical crown tree taxa, including Acer platanoides 'Globosum', Catalpa bignonioides 'Nana', Celtis occidentalis 'Globosa', Fraxinus excelsior 'Nana', Fraxinus ornus 'Mecsek', Platanus × hispanica 'Alphen's Globe', Prunus × eminens 'Umbraculifera', and Robinia pseudoacacia 'Umbraculifera'. A total of 4,676 specimens were assessed across Budapest's 23 districts, revealing significant variations in spatial distribution, with some districts having minimal or no presence of these species. [9]
2024	J. Sumberg M. Burke	Forest Tree Health Analysis	a methodology for classifying tree species and assessing their health status using unmanned aerial systems (UAS) equipped with multispectral imaging sensors. The research was conducted in a mixed broadleaf-conifer forest near Oplany, Czech Republic, covering a 180 × 160 m area dominated by Norway spruce, Scots pine, Downy oak, Hornbeam, and Silver birch. [?]

2024	NORHANE CHOUTER 1, MALIKA RACHED- KANOUNI2, OUARDA EL WAHIDA BOUCIF1	ASSESSMENT OF THE HEALTH STATUS OF OAK TREES IN THE ZOUAGHA FOREST	assessed the health status of oak trees in Algeria’s Zouagha Forest, focusing on three species: Quercus suber, Q. canariensis, and Q. afares. The methodology involved visual evaluations of tree crowns using the ARCHI, DEPEFEU, and IBP protocols. Field observations were conducted to apply these methods to individual trees of each species. The ARCHI protocol categorized trees into types such as Resilient (R) and Healthy (S), while the DEPEFEU protocol assessed health condition classes. The IBP method estimated the biodiversity carrying capacity of the forest. [?]
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III. PROPOSED METHOD

The proposed Tree Health Analyzer is designed to assess tree health using Java-based technologies by integrating image processing, machine learning, and IoT-based monitoring. This system aims to detect tree diseases, stress conditions, and environmental impacts in real-time, assisting foresters, researchers, and environmentalists in making informed decisions. The data acquisition module collects images of trees using drones, mobile cameras, or stationary surveillance cameras. Additionally, IoT sensors are deployed to monitor environmental parameters such as soil moisture, temperature, humidity, and chlorophyll levels. Java libraries such as JavaCV and OpenCV are used for image

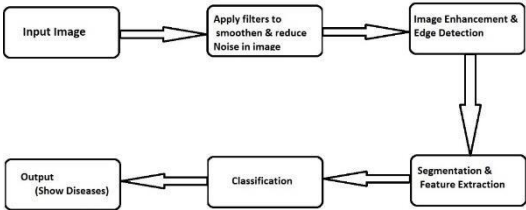
creation of RESTful APIs for seamless communication between the frontend and backend. The backend employs OpenCV (Java) for image preprocessing and feature extraction, while machine learning models are implemented using DeepLearning4J and Weka for classifying tree diseases based on extracted features. Additionally, Apache Spark MLlib is utilized for scalable data processing and predictive modeling. Sensor data collected from IoT devices, such as soil moisture and temperature sensors, is processed in real-time using MQTT and Java WebSockets. The backend also manages cloud storage and database interactions using MySQL or Firebase, ensuring long-term monitoring of tree health conditions. Cloud-based services like AWS S3 or

data is also normalized to maintain consistency across different devices and environmental conditions. In the feature extraction stage, relevant characteristics such as leaf color, texture, and shape are extracted using Java-based OpenCV and DeepLearning4J. Machine learning models are trained using Java frameworks like Weka and Apache Spark MLlib to classify tree diseases and health conditions accurately. Deep learning techniques, specifically Convolutional Neural Networks (CNNs), are implemented to enhance disease detection, providing a more precise assessment of tree health.

[3]

preprocessing, including noise reduction and contrast adjustment, ensuring high-quality input data. Sensor data is also normalized to maintain consistency across different devices and environmental conditions. In the feature extraction stage, relevant characteristics such as leaf color, texture, and shape are extracted using Java-based OpenCV and DeepLearning4J. Machine learning models are trained using Java frameworks like Weka and Apache Spark MLlib to classify tree diseases and health conditions accurately. Deep learning techniques, specifically Convolutional Neural Networks (CNNs), are implemented to enhance disease detection, providing a more precise assessment of tree health.

IV. FLOW CHART OF RELATED WORK and classification



Google Cloud Storage are integrated for scalable storage solutions.[7] notifications for early disease detection. The backend layer is responsible for handling data processing, machine learning-based disease detection, and database management. It is developed using Spring Boot, which facilitates

1. Data Collection

The system gathers data from various sources, including: User Input: Users scans parameters like leaf color, bark texture.

2. Image Processing

Converts the image to grayscale to simplify analysis. Applies filtering techniques (e.g., Gaussian blur) to remove unwanted

noise. Uses methods like Sobel or Canny edge detection to highlight leaf boundaries.

3.Feature Extraction Payment System

Detects discoloration (e.g., yellowing, brown spots) using HSV or RGB values. Identifies rough, dry, or decayed areas using contrast and pattern detection.

4.Disease Identification (Rule-Based Approach)

Symptom-Based Analysis – The system collects user-input. Diagnosis and Treatment Suggestion – Based on identified symptoms, the system suggests possible diseases.

5.Disease Report Recommendations

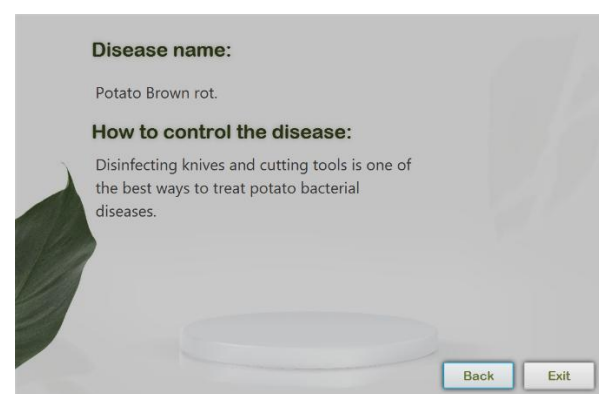
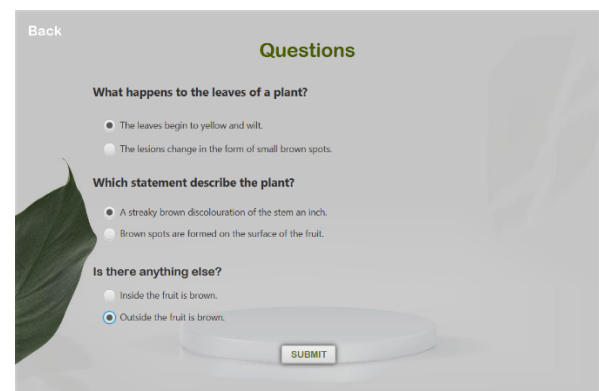
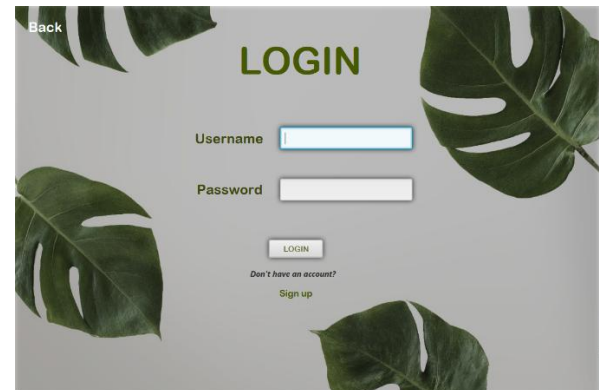
The system displays the identified disease. Provides remedial suggestions. [2]

V. RESULTS AND DISCUSSIONS

A. Implementation

The Tree Health Care Analyzer can be implemented using various Java technologies depending on the complexity and scalability requirements. A basic implementation can be done using Core Java with OOP principles, where a console-based system takes user input and analyzes tree health using conditional statements. For a more interactive graphical user interface (GUI), technologies like Java Swing or JavaFX can be used to create a user-friendly application with input fields and visual health indicators. If a web-based application is required, Spring Boot can be used to develop a RESTful API that interacts with a database such as MySQL, PostgreSQL, or MongoDB, allowing users to store and retrieve tree health data through a web interface built with Thymeleaf, React, or Angular. For advanced AI-driven analysis, Java-based machine learning frameworks like Deeplearning4j, Weka, or TensorFlow for Java can be integrated to predict tree health based on training data. The choice of technology depends on whether the application is meant for simple use, large-scale monitoring, or AI-based diagnostics. Furthermore, while Java provides a robust framework for image processing and system integration, incorporating machine learning models could significantly improve the accuracy and adaptability of the analyzer. Future enhancements could include integrating cloud-based data storage for continuous learning, developing a mobile application for on-the-go tree health assessment, and refining the algorithm to improve performance under diverse environmental conditions. Despite these challenges, the Tree Health Analyzer demonstrates significant potential in aiding environmental conservation, forestry management, and agricultural sustainability by providing an automated and efficient approach to tree health monitoring.[9]

B .Results



VI. CONCLUSION

The Tree Health Analyser is a Java-based application designed to scan leaves and identify diseases using image processing techniques without AI. By utilizing image prepro- cessing, feature extraction, and a rule-based decision system, the project provides an efficient and user-friendly solution for tree health monitoring. This application offers fast, reliable, and accessible disease detection for farmers, gardeners, and researchers. Its structured methodology—capturing images, analyzing leaf characteristics, and comparing them with pre- defined disease symptoms—ensures accurate results without requiring complex AI models. Additionally, the project is scalable and customizable, allowing future enhancements such as integrating sensor data, expanding the disease database, or developing a mobilefriendly version. With a well-designed Java-based GUI, users can easily interact with the System, view results, and get treatment recommendations.[5

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